

DepChain: Stage 1 Report

João Gouveia, Gonçalo Rua, and Francisco Fonseca

Instituto Superior Técnico

1 Introduction

DepChain is a permissioned blockchain system designed with a focus on dependability. This report details the first stage of its development, emphasizing the consensus layer based on the Byzantine Read/Write Epoch Consensus algorithm presented on p. 252 of *Introduction to Reliable and Secure Distributed Programming*. As stated in the guidelines, we implemented the Byzantine Read/Write Epoch Consensus with the following simplifying assumptions:

- The system membership is static for the entire system lifetime, including a pre-defined leader process. We always assume $\text{pid}=0$ is the leader process.
- There is a Public Key Infrastructure in place.
- There's no leader change.

2 Design and Justification

As stated in the guidelines, the following assumptions were followed:

- The application can trust its local instance of the client library.
- A subset of the blockchain members may be malicious and behave in an arbitrary manner.
- The network is unreliable: it can drop, delay, duplicate, or corrupt messages, and communication channels are not secured.

The DepChain system consists of two main components, a **client library** that translates application calls into requests for the blockchain service and the **blockchain members** that store the state and execute the consensus protocol.

The state stored by each blockchain member, what we called EpochState, includes the current timestamp, the current value and a writeset of timestamps and values, the values being all those the member has ever written, with the corresponding timestamp denoting the most recent epoch in which it was written - i.e. if a process p has a writeset = $((3, "A"), (5, "B"))$ it means p wrote "A" at timestamp 3 and "B" at timestamp 5.

The Byzantine Read/Write Epoch consensus algorithm is running in each process and, as its name implies, it has two different phases. The Read phase begins with the initialization of the EpochState with current timestamp and value and the initialization of two empty lists, written and accepted.

We used the Signed Conditional Collective primitive in our implementation as this enables the use of the primitive with byzantine processes. Every time the

Conditional Collect receives messages from enough processes, it adds a digital signature corresponding to each message and then sends this signature along with the message to the leader. The leader then verifies if the signatures match with their corresponding message and broadcasts the correct messages and signatures to all processes.

When the leader decides to propose a new value, it broadcasts to all other members that a new value is to be written into the blockchain. But before the value the leader wants to propose can be written, in order to guarantee consistency between all members, everyone sends their current EpochState to the leader, through the Signed Conditional Collect primitive, which will then search for a value that may have been previously decided by a process. If such value is found, it is signaled to all members to be written.

All communication is done through Authenticated Perfect Links, which besides reliable delivery and no duplication, ensure a message was not tampered with between its sender and receiver.

3 Dependability Guarantees

In this stage, the system is designed to provide: (1) Safety, (2) Liveness under a correct leader and (3) Integrity.

- **Safety** is the property which states that no catastrophic event will happen to any user or to the environment. This is achieved by the consensus algorithm which ensures that all correct blockchain members reach a consistent state, and even if some members behave maliciously, as byzantine faults can do, no conflicting decisions can be made - the blockchain remains accurate. This is achieved through the algorithm which uses quorums in order to filter out malicious actions by byzantine members.
- **Liveness** means the system will eventually progress, as long as the leader is correct, meaning as long as there's a well functioning leader communicating with the other processes about what they should do next, each process will know what to do and the continuation of the program is guaranteed. As there's no leader election at this stage, there's no way to guarantee liveness under a byzantine leader, something that will change on the next stage.
- **Integrity** simply means the absence of undesirable system alterations, something attainable by the use of the Authenticated Perfect Links which enable to understand if a message was tampered with by someone other than its sender.

4 Conclusion

This stage of the DepChain project establishes the basis for the blockchain's consensus layer. Future work will extend the system to transaction processing, implementation of a leader election process, which improves Byzantine fault tolerance, and persistent storage for the blockchain.